

RISHABH RASTOGI

Ph.D. · Senior Process Engineer · ALD | PVD | Sub-3nm GAA & CFET Gate Stack
Gurugram, India • rishabhrastogi.phd@gmail.com • +91-99998 87765 • linkedin.com/in/rishabh-rastogi-phd •
rishabhrastogiphd.netlify.app

PROFESSIONAL SUMMARY

PhD materials scientist with 8+ years of thin-film work in ALD and PVD: about 4 years of doctoral research at LIST (a research and technology organisation), nearly 3 years of 300mm industrial R&D with ASM International (IMEC, Leuven, then Arizona), and independent consulting since 2025. Core area is FEOL high-k/metal gate stack module development for sub-3nm GAA and CFET: interface dipole engineering, capping layers, gate dielectrics, interfacial layer doping, and gapfill. Work covered recipe development, tool operation, metrology, electrical validation, and process transfer to customer fabs. Co-inventor on 7 patent filings; 6 peer-reviewed publications. Open to senior process engineering, applications engineering, and R&D roles globally, including roles with significant customer travel.

CORE TECHNICAL SKILLS

Deposition: ALD (thermal; PEALD at operational level) · PVD (sputter, e-beam, thermal evaporation) · MLD · Gapfill & conformal deposition · 300mm wafer processing · 300mm wet chemical etch

Gate Stack Module: FEOL gate stack module development (sub-3nm GAA nanosheet, CFET) · Interface dipole engineering (dipole-first, dipole-last) · Capping layer optimisation · Gate dielectrics & interfacial layer doping · eWF/Vt modulation with EOT & Jg control · FD-SOI · Process transfer & qualification (IQ/OQ/PQ)

Electrical: In-house MOSCAP device flow · C-V / I-V measurement · Flat band voltage extraction

Metrology & Characterization: XPS · SIMS · TEM/EDS · SEM (incl. CD-SEM) · Cross-sectional TEM/SIMS/XPS in X-cut & Y-cut on GAA · AFM · XRD/XRR · Ellipsometry · Profilometry · Resistivity · Defect inspection (SP5-class) · FDTD simulation

Process & Methods: Recipe development & optimisation · DOE (JMP) · SPC · Tool maintenance & troubleshooting · Six Sigma Green Belt · Technical reporting & cross-functional communication

PROFESSIONAL EXPERIENCE

Independent Consultant — Semiconductor Process Technology | Freelance

Mar 2025 – Present • Remote, India

- Expert advisory for PE firms, hedge funds, and asset managers through institutional expert networks. Topics include High-NA EUV, advanced lithography, and the competitive positioning of ASM International and TEL.
- Part-time technical advisor to a Switzerland-based startup planning a 28nm logic fab. Scoped process architecture, equipment requirements, and resource plan. Engagement ended due to client-side funding constraints.
- ALD process feasibility assessments and equipment benchmarking for OEM and deep tech clients. Evaluate tool performance, process capability, and film quality; advise on integration options for logic and memory flows.
- Guest lectures and workshops at universities in India and abroad on semiconductor manufacturing, ALD, and deep tech careers.

Senior Process Engineer | ASM International N.V.

Jun 2022 – Mar 2025 • Leuven, Belgium, embedded at IMEC 300mm fab (to Jul 2023) → Arizona, USA

ALD process development for sub-3nm GAA and CFET gate stack modules. First year embedded at IMEC's 300mm research foundry, then ASM's Arizona business unit on later-stage, customer-facing programmes. Owned deposition modules; co-designed DOEs with device and integration teams, who owned the full device flow.

- Worked with 15+ ALD chemistries for gate stack layers: rare-earth-based dipole and capping layers, high-k dielectrics, and interfacial layers. Scope ranged from optimising an existing POR to screening new precursors for a stable ALD window and GPC recipe.
- Interface dipole engineering in dipole-first and dipole-last schemes. Tuned deposition, etch, and drive-in conditions to shift eWF and Vt with negligible or improved EOT and Jg.

- Validated film changes electrically through an in-house MOSCAP flow, extracting flat band voltage shifts from C-V and I-V data.
- Ran 300mm wet chemical etch steps within the FEOL flow, supporting gate stack and gapfill work.
- Developed ALD gapfill for sub-1nm lateral gaps in GAA nanosheet structures (patent pending). Assessed fill, conformality, and voids by SIMS, TEM, and SEM.
- Metrology: XPS, SIMS, TEM/EDS, SEM, and AFM for composition, interfaces, and morphology; ellipsometry and XRR for thickness; SP5-class brightfield inspection for defects. Cross-sectional TEM, SIMS, and XPS in X-cut and Y-cut orientations on GAA structures.
- Operated and troubleshoot 300mm thermal ALD platforms. Tracked tool health through film metrics, coordinated maintenance and calibration with hardware teams, and caught process drift early.
- Designed DOE matrices and applied SPC to isolate process parameters and keep film quality reproducible across tools.
- Supported 4+ process transfers from alpha development to HVM beta qualification at customer fabs: documentation, on-site support, and troubleshooting of deviations.
- Worked with customer engineering teams on sub-3nm device development, diagnosing thin-film integration issues in GAA flows.
- Ran knowledge-sharing seminars for US and Belgium R&D teams on ALD process development, film scaling, and tool performance.

Research Engineer (Doctoral) | Luxembourg Institute of Science and Technology (LIST)

Nov 2016 – Nov 2020 • Luxembourg / University of Technology of Troyes, France

- Fabricated Si nanostructured plasmonic arrays at wafer level using sputtering, e-beam and thermal evaporation, ALD, and lithography. Controlled film thickness, morphology, and nanostructure geometry.
- Engineered thin-film interfaces in sub-10nm gap structures to localise electromagnetic fields. Characterised by XPS, spectroscopic metrology, and FDTD simulation.
- Developed a QCM-plasmonic sensing interface with picomolar detection sensitivity (patent filed, 2019).
- Operated and maintained deposition and characterisation tools in a multi-user research environment.

Co-Founder | Flamboyant Technologies & SanWalk

Feb 2021 – May 2022 • India

- Flamboyant Technologies: Business development and solution scoping for ERP implementations (MS Dynamics 365) across 10+ enterprise clients.
- SanWalk: Led product concept, MVP design, and early-stage prototyping for a consumer hardware venture.

Associate Engineer | Intas Pharmaceuticals

Jan 2016 – Jul 2016 • India

- Analytical method development, validation, and transfer for biopharmaceuticals in a GMP-regulated environment.

Visiting International Researcher | University of South Australia

Jan 2015 – Jul 2015 • Australia — Fully Paid Master's Thesis Research (Amity B.Tech + M.Tech programme)

- Developed porous silicon nanocarriers by nanosphere lithography and electrochemical etching, for drug delivery with microwave-triggered thermal release.

EDUCATION

Ph.D. — Materials Science and Nanoscale Engineering | University of Technology of Troyes, France | 2016–2020

B.Tech + M.Tech — Nanotechnology (Devices) | Amity University, Noida, India | 2010–2015

PATENTS & PUBLICATIONS

7 patent filings (co-inventor): gate stack dipoles, interfacial layer doping, threshold voltage control, and gapfill at ASM International (2024–2025); plasmonic sensing at LIST (2019) · 6 peer-reviewed journal papers (4 first-author; ACS Applied Materials & Interfaces, Journal of Physical Chemistry C, Sensors and Actuators) · 2 book chapters · 7 conference presentations

Full patents and publications list in Appendix.

APPENDIX — FULL PATENTS & PUBLICATIONS

Rishabh Rastogi, Ph.D.

PATENTS (7 FILED, CO-INVENTOR)

- ASM International patent applications (2024–2025): interfacial layer doping, rare-earth dipole layers, threshold voltage control, and substrate processing. Titles and application numbers available on request.
- R. Rastogi et al., "Quartz Crystal Microbalance with Plasmonic Sensing Capacity." Filed Aug 2019. LU101354 | WO2021032744.

Further ASM filings in progress (gate stack dipoles, gapfill, patterning sacrificial materials).

JOURNAL PUBLICATIONS

- R. Rastogi et al., "Analyte Co-localization at Electromagnetic Gap Hot-Spots for Highly Sensitive Molecular Detection by Plasmon Enhanced Spectroscopies." ACS Applied Materials & Interfaces, 2021, 13(7), 9113–9122. DOI: 10.1021/acsami.0c17929.
- R. Rastogi et al., "Engineering Electromagnetic Hot-Spots in Nanoparticle Cluster Arrays on Reflective Substrates." ACS Applied Materials & Interfaces, 2021, 13(28), 32653–32661. DOI: 10.1021/acsami.1c01953.
- R. Rastogi et al., "Hierarchically Structured Plasmonic Nanoparticle Assemblies with Dual-Length Scale EM Hot Spots." Journal of Physical Chemistry C, 2021, 125(16), 8647–8655.
- R. Rastogi et al., "Quantifying Analyte Surface Densities with Respect to Electromagnetic Hot Spots in Plasmonic Biosensors." Journal of Physical Chemistry C, 2021, 125(18), 9866–9874.
- M. Beggiano, R. Rastogi et al., "Confined Adsorption within Nanopatterns for High Adsorption Efficiencies on Affinity Sensors." Sensors and Actuators B, 2022, Vol. 366.
- C.B.A. Stoffels, R. Rastogi et al., "Plasmonic Microcarriers for Sensing and Cell Expansion." Sensors and Actuators Reports, 2023, Vol. 6.

BOOK CHAPTERS & THESIS

- R. Rastogi et al., "Structure Transfer by Lithography" in Nanomaterials for Nano Spectroscopy (Ch. 6). De Gruyter.
- R. Rastogi et al., "Nanoplasmonic Arrays for Quantitative Molecular Detection" in Plasmonics. IntechOpen, 2019.
- PhD Thesis: "Engineered Electromagnetic Hot-spots for Highly Sensitive Molecular Detection by Plasmonic Spectroscopies." University of Technology of Troyes, 2020. Ref: 2020TROY0018.

SELECTED CONFERENCE PRESENTATIONS

- "Wafer Level Fabrication of sub-10nm Gap Hot Spots..." — ICES, Ontario, Canada, 2019 (Invited Oral).
- "Engineering Nanopattern Geometry for Nanoplasmonic Biosensor Sensitivity." — SPIE, 2018 (Oral).
- "Controlling Nanoscale Geometries for High-Performance Plasmonic Biosensing." — NFO 2018 (Poster).
- 4+ additional presentations (2017–2019).